

\$ Disk Utilities

Linux Command Line — Session 2

Picking up where Process Utilities left off

`du`

`df`

`mount`

`umount`

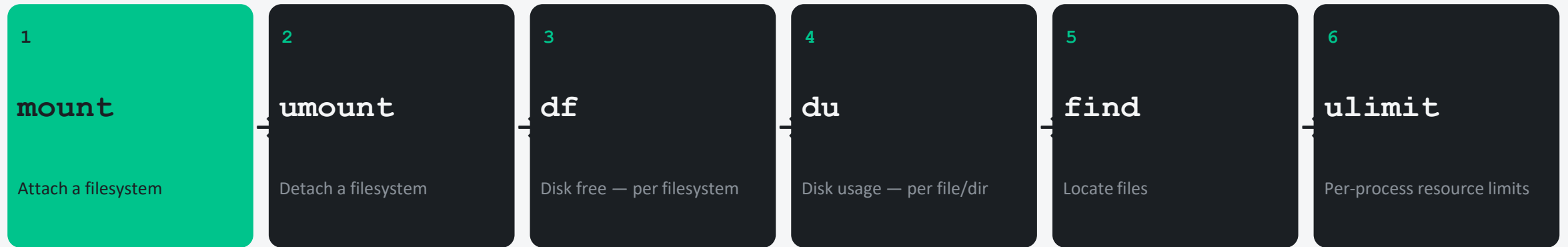
`find`

`umask`

`ulimit`

Today's Flow

Six commands, in the order we'll cover them



We just finished Process Utilities (ps, background & foreground jobs) — now we shift from managing processes to managing storage.



mount

Attach a filesystem to the directory tree

SYNTAX

```
mount [-t type] device dir
```

KEY POINTS

- Makes a device or partition's contents accessible at a mount point
- `-t` — specify filesystem type (ext4, ntfs, iso9660...)
- Run with no arguments to list all currently mounted filesystems
- Mount point must already exist as an empty directory
- Typically requires root / sudo privileges

EXAMPLE

```
$ sudo mount -t ext4 /dev/sdb1 /mnt/data
```

Mounts partition sdb1 at /mnt/data so its files appear under that path.



umount

Detach a mounted filesystem safely

SYNTAX

```
umount dir | device
```

KEY POINTS

- Reverses mount — cleanly unlinks the filesystem from the tree
- Can reference either the mount point or the device name
- “device busy” error → something still has a file open there
- Use lsof or fuser -m <dir> to find what's holding it open
- Always unmount removable media before physically disconnecting

EXAMPLE

```
$ sudo umount /mnt/data
```

If this fails with 'target is busy', check open files before forcing anything.



df

Disk Free — space available on mounted filesystems

SYNTAX

```
df [options]
```

KEY POINTS

- Reports total, used, and available space per mounted filesystem
- -h — human-readable sizes
- -T — show filesystem type (ext4, xfs, tmpfs, etc.)
- Operates at the filesystem/partition level, not per-file
- First command to check when you get a “no space left” error

EXAMPLE

```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       50G   32G   18G   64% /
```

Now that mount points are familiar, 'Mounted on' in this output should click right away.



du

Disk Usage — how much space files & directories consume

SYNTAX

```
du [options] [path]
```

KEY POINTS

- Reports space used by files and directories, recursively
- `-h` — human-readable sizes (K, M, G)
- `-s` — summarize (total only, not every subfolder)
- `--max-depth=N` — limit how deep the breakdown goes
- Useful for finding what's eating up space inside a filesystem

EXAMPLE

```
$ du -sh /var/log  
2.3G    /var/log
```

df looked at the whole filesystem; du zooms in to files and directories.



find

Search the filesystem for files matching criteria

SYNTAX

```
find [path] [expression]
```

KEY POINTS

- Recursively searches a directory tree for matching files
- `-name` — match by filename pattern
- `-type f` / `-type d` — restrict to files or directories
- `-mtime`, `-size` — filter by modification time or size
- `-exec` — run a command on every match found

EXAMPLE

```
$ find /home -name "*.log" -mtime +7
```

Finds .log files under /home not modified in the last 7 days.



ulimit

Per-process resource limits for the current shell

SYNTAX

```
ulimit [-S|-H] [option]  
[value]
```

KEY POINTS

- Controls resource ceilings for processes started from the shell
- -a — display all current limits
- -n — max open file descriptors
- -u — max number of user processes
- -S (soft) vs -H (hard) — soft can be raised up to the hard limit

EXAMPLE

```
$ ulimit -n  
1024  
$ ulimit -n 4096
```

Raises the open-file-descriptor limit for this shell session.

Recap

`mount / umount`

Attach and detach filesystems from the tree

`df / du`

Measure space — per filesystem vs. per file

`find`

Locate files by name, type, time, or size

`ulimit`

Per-process resource caps for the shell

Next up: putting it together with real disk-troubleshooting exercises.